

ROMA TRE · A.Y. 2026/27

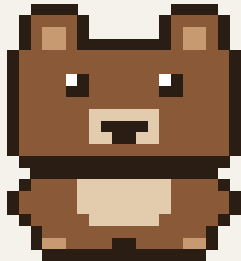
Policy Evaluation Lab

Lecture 4: Social experiments (I)

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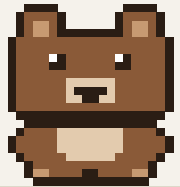
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PART 1

Randomised experiments



Approaches to identify causal effects

- The main challenge in evaluating a public policy, and the goal of most empirical economic research, is the elimination of **selection bias**
- ...and to say something about the **causal effect**
- The best approach is based on a **randomised experiment**, also called a randomised controlled trial (**RCT**)



Randomisation

- **Social experiments:** individuals eligible for the programme, and who agree to take part in the experiment, are divided into two random groups
 - a **treatment group**, which benefits from the programme, and a **control group**, which does not
- Assignment to treatment is *random* in the population

Randomisation ensures that the sample selection bias is zero and the treatment effect is **identified**



How does it work?

Start with a sample of uninsured individuals.

- Suppose we have two, Karl and Jenny
- We toss a coin: if it comes up heads, Karl gets the coverage, otherwise it goes to Jenny
- We have random assignment, but it is not enough: Karl and Jenny are different (e.g. in gender)
- Women are healthier than men: if Jenny gets insured and is healthier,
- it may be due to the luck of being born a woman, and unrelated to her lucky draw in the lottery
- Hence, we need a **large sample** to ensure that differences in individual characteristics are eliminated



PART 2

Random assignment



Law of Large Numbers

- By increasing the sample size, we can ensure that the sample average gets as close as we like to the population average (**Law of Large Numbers**, LLN)
- By randomly assigning individuals from the same population to the treatment and control groups, we create two groups in a way that is similar to a repeated coin toss
- Therefore, we will have two groups whose average characteristics are close to those of the population
- This is true only when the groups are large enough



No selection bias

- These groups would be similar in every possible way, even in terms of **unobserved** characteristics
- E.g. we expect similar proportions of men and women, and similar age and schooling levels
- Randomisation does not eliminate individual differences (it does not transform an apple into an orange)
- It ensures that the two groups contain the same mix of individuals (the two barrels contain the same proportion of apples and oranges)
- The treated and control groups differ only in their **treatment status** and its consequences

Experimental randomisation eliminates **sample selection bias**



Random assignment eliminates selection bias

If the sample is large enough for the LLN to work, we can replace the conditional averages with conditional expectations. With a constant treatment effect, $Y_{1i} = Y_{0i} + \tau$:

$$\begin{aligned} E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] &= E[Y_{1i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0] \\ &= E[Y_{0i} + \tau \mid D_i = 1] - E[Y_{0i} \mid D_i = 0] \\ &= \tau + E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0] \\ &= \tau \end{aligned}$$

- Randomisation of D_i ensures that $E[Y_{0i} \mid D_i = 1] = E[Y_{0i} \mid D_i = 0]$,
- because randomly assigned treatment and control groups come from the same population, they are the same in every way, including Y_{0i}



Balance check

- The first step when trying to estimate a causal effect is **checking for balance** between the treated and control groups
- This implies comparing the average characteristics of treated and controls
- Random assignment is the best way to guarantee that such balance is achieved

Remember Panel B of Table 2 (NHIS data, Lecture 2): the data do not come from an experiment, and the average characteristics are dissimilar.



PART 3

Application: the Oregon Health Plan



Oregon Trail (I)

- A new experiment to check the effect of the **Medicaid** expansion
- Medicaid covers families on welfare, some disabled people, poor children and poor pregnant women
- The State of Oregon offered Medicaid (the **Oregon Health Plan**, OHP) to randomly chosen people in previously ineligible groups
- Probably the best evidence on the effects of health insurance (HI) on health costs and outcomes



Oregon Trail (II)

- Lottery winners were randomly selected from a pool of registrants
- They had the opportunity to apply for the OHP, but they still had to demonstrate they were poor
 - The state reviewed applications, awarding coverage to people aged 19–64 with no health coverage and income below the poverty level
- 75,000 lottery applicants: 30,000 were selected to apply for the OHP (**treatment group**), and 45,000 constituted the **control group**



Were lottery winners more likely to get Medicaid? (I)

Outcome	Oregon		Portland area	
	Control mean (1)	Treatment effect (1)	Control mean (2)	Treatment effect (3)
<i>A. Administrative data</i>				
Ever on Medicaid	.141	.256 (.004)	.151	.247 (.006)
Any hospital admissions	.067	.005 (.002)		
Any emergency department visit			.345	.017 (.006)
Number of emergency department visits			1.02	.101 (.029)
Sample size	74,922		24,646	



Were lottery winners more likely to get Medicaid? (II)

Outcome	Oregon		Portland area	
	Control mean (1)	Treatment effect (1)	Control mean (2)	Treatment effect (3)
<i>B. Survey data</i>				
Outpatient visits (in the past 6 months)	1.91	.314 (.054)		
Any prescriptions?	.637	.025 (.008)		
Sample size	23,741			



OHP effects on coverage and health-care use

- 14% of lottery losers were covered by Medicaid in the year after the OHP lottery
- However, the probability of Medicaid coverage increased by 26 percentage points in the treatment group
- Hospital admissions increased slightly
- Emergency department visits increased too (perhaps counterintuitively)



OHP effects on health (I)

Outcome	Oregon		Portland area	
	Control mean (1)	Treatment effect (2)	Control mean (3)	Treatment effect (4)
<i>A. Health indicators</i>				
Health is good	.548	.039 (.008)		
Physical health index			45.5	.29 (.21)
Mental health index			44.4	.47 (.24)
Cholesterol			204	.53 (.69)
Systolic blood pressure (mm Hg)			119	-.13 (.30)



OHP effects on health (II)

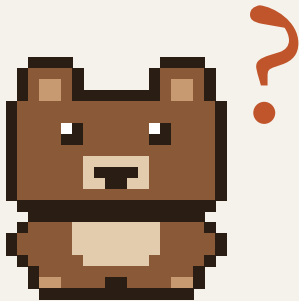
Outcome	Oregon		Portland area	
	Control mean (1)	Treatment effect (2)	Control mean (3)	Treatment effect (4)
<i>B. Financial health</i>				
Medical expenditures >30% of income			.055	-.011 (.005)
Any medical debt?			.568	-.032 (.010)
Sample size	23,741		12,229	



OHP conclusions

- Lottery winners have a higher probability of reporting good health (+0.039)
- This improvement derives from improved mental health (+0.47)
- Physical health indicators are basically unchanged
- A disappointing result for policymakers: the OHP did not generate a health dividend and increased emergency department use
- It provided a **financial safety net** for the poor
- However, these effects are driven by the roughly 25% of winners who obtained HI because of the lottery (i.e. changed status)
- Adjusting for the fact that insurance status was unchanged for many winners (e.g. some were already on Medicaid) means that the gains for those who did get insured were actually much larger





Questions?

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